

Society are mentioned below. Two of them are incomplete drafts in Newton's handwriting, and made, I conjecture, prior to the memoir being sent to London. The third, probably also holograph, is carefully written out, and is somewhat fuller than that in the archives of the Royal Society; it may have been written subsequently to the copy sent to London, but I think it is of a date not later than the spring of 1685; at any rate, it contains a remark (see below, p. 56) which implies that though Newton knew that, at a considerable distance from the earth's centre, gravity might be taken for purposes of calculation to vary inversely as the square of the distance from that point, yet he was then uncertain whether the same law held for points near the earth's surface. As, however, the date and history of the Royal Society's copy are accurately known, I have printed the tract in the form there written except in three respects: in the first place I have corrected the diagrams, and all purely clerical blunders, by Newton's own copy; in the second place I have inserted punctuation, and in some cases have altered the initial letters of words to capitals or small letters as the case may be, so as to make the practice uniform (in these respects Rigaud has not followed the manuscript exactly); and in the third place I have written the symbol $\times$ instead of $+$, and instead of *et*, when $+$ and *et* are used (as in these manuscripts they not infrequently are) to indicate multiplication. I have also replaced any symbol like ABC by $AB \times BC$ when it is used to denote the product of AB and BC . I need hardly add that in the writings of mathematicians of the seventeenth century expressions like x^a , x^c , x^{aq} , x^{qc} stand for what we usually write as x^2 , x^3 , x^4 , x^5 .

The tract is as follows:

ISAACI NEWTONI PROPOSITIONES DE MOTU.

Definitiones.

1. VIM centripetam appello, qua corpus impellitur vel attrahitur versus aliquod punctum, quod ut centrum spectatur.
2. Et vim corporis, seu corpori insitam, qua id conatur perseverare in motu suo secundum lineam rectam.
3. Et resistantiam, quae est medii regulariter impredientis.

Hypotheses.

1. Resistentiam in proximis novem propositionibus nullam esse, in sequentibus esse ut medii densitas et celeritas conjunctim.

2. Corpus omne sola vi insita uniformiter secundum lineam rectam in infinitum progredi, nisi aliquid extrinsecus impediat.

3. Corpus in dato tempore, viribus conjunctis, eo ferri quo viribus divis in temporibus aequalibus successive.

4. Spatium quod corpus, urgente quacunque vi centripeta, ipso motus initio describit, esse in duplicata ratione temporis.

Lemmata.

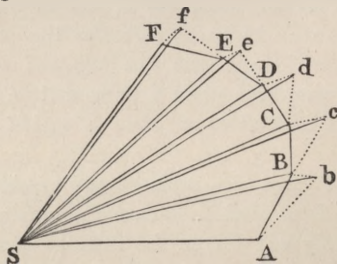
1. Parallelogramma omnia, circa datam ellipsin descripta, esse aequalia inter se. Patet ex conicis.

2. Quantitates differentiis suis proportionales sunt continue proportionales. Ponatur A ad $A-B$ ut B ad $B-C$, et C ad $C-D$, &c. dividendo fiet A ad B , ut B ad C , ut C ad D , &c.

Propositiones.

Theor. I. Gyratione omnia radiis ad centrum ductis areas temporibus proportionales describere.

Dividatur tempus in partes aequales, et prima temporis parte describat corpus vi insita rectam AB ; idem secunda temporis parte, si nihil impedit, recta pergeret ad c , describens rectam Bc aequalem ipsi AB ,

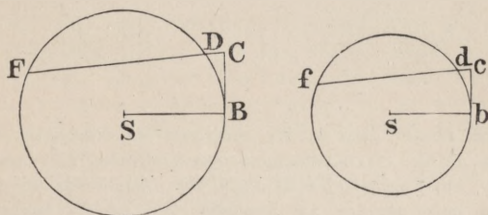


adeo ut radii AS , BS , CS ad centrum actis, confectae forent areae aequales ASB , BSc . Verum ubi corpus venit ad B , agat vis centripeta impulsu unico at magno, faciatque corpus a recta Bc deflectere et pergere in recta BC . Ipsi BS parallela agatur cC occurrens BC in C , et, completa secunda temporis parte, corpus reperietur in C . Junge SC , et triangulum SBC ob parallelas SB , Cc aequale erit triangulo Sbc , atque adeo etiam triangulo SAB . Simili argumento, si vis centripeta successive agat in C , D , E , &c. faciens corpus singulis temporis momentis singulas describere rectas CD , DE , EF , &c., triangulum SDC triangulo SBC , et SDE ipsi SCD , et SEF ipsi SDE aequale erit. Aequalibus

igitur temporibus aequales areae describuntur. Sunt jam haec triangula numero infinita, et infinite parva, sic ut singulis temporis momentis singula respondeant triangula, cogente vi centripeta sine remissione, et constabit propositio.

Theor. II. Corporibus in circumferentiis circularum uniformiter gyrantibus, vires centripetas esse ut arcuum simul descriptorum quadrata applicata ad radios circularum.

Corpora B, b , in circumferentiis circularum BD, bd gyrantia, simul describant arcus BD, bd . Sola vi insita describerent tangentes BC, bc ,



his arcubus aequales; vires centripetae sunt quae perpetuo retrahunt corpora de tangentibus ad circumferentias, atque adeo hae sunt ad invicem ut spatia ipsis superata CD, cd ; id est productis CD, cd ad F et f ut $\frac{BC^{\text{quad}}}{CF}$ ad $\frac{bc^{\text{quad}}}{cf}$, sive ut $\frac{BD^{\text{quad}}}{\frac{1}{2}CF}$ ad $\frac{bd^{\text{quad}}}{\frac{1}{2}cf}$. Loquor de spatiis BD, bd minutissimus, inque infinitum diminuendis, sic ut pro $\frac{1}{2}CF, \frac{1}{2}cf$, scribere liceat circularum radios SB, sb , quo facto constabit propositio.

Cor. 1. Vires centripetae sunt ut celeritatum quadrata applicata ad radios.

Cor. 2. Et reciproce ut quadrata temporum periodicorum applicata ad radios.

Cor. 3. Unde si quadrata temporum periodicorum sunt ut radii circularum, vires centripetae sunt aequales; et vice versa.

Cor. 4. Si quadrata temporum periodicorum sunt ut quadrata radiorum, vires centripetae sunt reciproce ut radii.

Cor. 5. Si quadrata temporum periodicorum sunt ut cubi radiorum, vires centripetae sunt reciproce ut quadrata radiorum.

Theor. III. Si corpus circa centrum S gyrando describat lineam quamvis curvam APQ ; et si tangat recta PR curvam illam in puncto quovis P , et ad tangentem ab alio quovis puncto Q agatur QR distantiae SP parallela, ac demittatur QT perpendicularis ad distantiam SP ; dico quod vis centripeta sit reciproce ut solidum $\frac{SP^{\text{quad}} \times QT^{\text{quad}}}{QR}$, si modo solidi illius ea semper sumatur quantitas, ubi coeunt puncta P et Q .